

APPROACH TO GASTROENTERITIS

IN PRIMARY CARE

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Definition

Gastroenteritis refers to inflammation in the stomach and intestines, most often resulting in diarrhoea with or without vomiting. Diarrhoea is usually defined as passage of abnormally liquid or unformed stools at an increased amount and frequency. An increase in frequency of bowel movement of 3 or more stools per day is generally used as a definition of diarrhoea for epidemiological studies. Diarrhoea is defined as acute if it lasts for 14 days or less. Chronic diarrhoea is diarrhoea that has lasted more than 30 days.¹

For the purpose of management, we can classify based on the following severity²:

Severity	Definition
Mild diarrhoea	- ≤ 3 stool movements/day - Diarrhoea is bearable, and the patient is capable of traveling or other activities as planned
Moderate diarrhoea	- $\geq 4 - 6$ stool movements/day - Diarrhoea interferes with planned travels or other activities.
Severe diarrhoea	- > 6 bowel movements/day or bloody diarrhoea - Diarrhoea interferes with daily activities and prevents planned travels or other activities.

Diagnosis

The diagnosis of gastroenteritis is clinical. The aim of evaluation is to distinguish patients with mild, self-limiting diarrhoea from those requiring further investigations, empirical antibiotic therapy or admission.

For diagnostic and management purposes, it is useful to classify acute diarrhoeas into “non-inflammatory” and “inflammatory” syndromes.¹

Diarrhoeal syndromes	Organisms commonly implicated
Non-inflammatory	Viruses: <i>Norovirus</i> , <i>rotavirus</i> Bacteria: <i>Clostridium perfringens</i> , <i>Staphylococcus aureus</i> , <i>Vibrio cholerae</i> Parasites: <i>Giardia lamblia</i> , <i>Entamoeba histolytica</i>
Inflammatory	<i>Shigella</i> , <i>Salmonella</i> , <i>entero-haemorrhagic Escherichia coli (EHEC)</i> , <i>enteroinvasive Escherichia coli (EIEC)</i> , <i>Campylobacter</i> , <i>Clostridium difficile</i>

1. Non-inflammatory diarrhoea syndrome

This is characterized by watery stools of large volume without blood. The patient can have nausea and vomiting, abdominal colic and low grade fever. In most cases, this diarrhoea is self-limiting and benign. Investigations are generally unnecessary unless cholera is suspected. Antibiotics are usually not required.

2. Inflammatory diarrhoea syndrome

This is characterized by frequent, small volume stools which may be bloody. It is often accompanied by fever, tenesmus and severe abdominal pain. This syndrome implies invasion or mucosal damage by the microbe or cytotoxins. Investigations and antibiotics may be required.

History

Pertinent history to obtain:

1. Age - Young children and elderly are more likely to suffer complications of dehydration and sepsis. The majority of mortality from gastroenteritis occurs in these extremes of age.
2. Characterize the symptoms - number of diarrhoea or vomiting, presence of watery or bloody stools
3. Ability to tolerate orally - Patients who are unable to keep fluids will be of a higher risk of dehydration
4. Past medical history - Patients with an immunocompromised state are of a higher risk of sepsis. Immunocompromised states include those on long-term steroids, immunosuppressive or chemotherapy, HIV, or chronic organ failure.
5. Travel history - Diarrhoea which occurred during travel is classified as traveller's diarrhoea. The expected causative organisms are those which are prevalent in the visited country.
6. Contact history - This is important from a public health point of view. Clinicians should be on high alert if there is a common source of infection and alert authorities to prevent further spread of the organism.
7. Previous use of antibiotics - This group of patients have to be investigated for nosocomial infections, especially *Clostridium difficile* infections.
8. Occupation - Food handlers should be advised to only return to work 48 hours after last diarrhoea or vomiting to prevent transmission of organisms

Other differentials should be considered when a patient presents with vomiting only without diarrhoea.

Examination

Perform a directed physical examination:

1. Vital signs: Record the temperature, blood pressure (including postural BP if needed), heart rate
2. Signs of dehydration: postural drop, hypotension, tachycardia, sunken fontanelles (for infants), loss of skin turgor, increased cap refill time, dry mucous membranes
3. Examine the abdomen for tenderness, distension, rigidity or guarding
4. Perform a per rectal examination to look for the presence of blood in stools if history is suggestive

Investigations

Most cases of gastroenteritis in the community do not require further evaluation and are self-limiting. A careful evaluation of a patient with a good history and physical examination can save unnecessary investigations and use of antibiotics.

Investigations are only required if the diarrhoea is prolonged or there is bloody diarrhoea. Stool culture for *Salmonella*, *Shigella* and *Campylobacter* etc. can be sent.

For patients who develop diarrhoea after hospitalization or recently received antibiotics, patients should be referred to tertiary settings for evaluation of *Clostridium difficile*.

Protozoal parasites are uncommon causes of traveller's diarrhoea but should be considered when diarrhoea lasts beyond a week or bacterial causes have been treated. Stool culture for ova, cysts and parasites (OCP) can be sent.

Treatment

Most cases of gastroenteritis in the community only require supportive treatment.

1. Rehydration

Fluid and electrolyte replacement is essential in the management of all patients with acute gastroenteritis. Oral rehydration is the treatment of choice.

If a patient can take orally, isotonic drinks or oral rehydration salts will suffice. Patients who are unable to tolerate orally may require intravenous hydration in the tertiary setting.

2. Anti-diarrhoea agents

Antidiarrheal agents include intestinal motility inhibitors, intestinal secretion inhibitors and absorbents. Intestinal secretion or motility inhibitors may be helpful in decreasing the frequency or duration of diarrhoea for symptomatic improvement of acute infectious diarrhoea patients with moderate symptoms.

	Effectiveness	Dose
Intestinal motility inhibitor <i>Antimotility agents are also not recommended in the management of acute gastroenteritis in infants and children because they do not address the underlying cause and have a risk of serious adverse events.⁶</i>		
Loperamide	Inhibits intestinal motility to decrease the movement of intestinal contents and promote the absorption, thereby decreasing diarrhoea. It also suppresses the secretion of intestinal mucosal secretion, which contributes to decreasing diarrhoea. It shortens the duration by 1 day, and decreases the amount and frequency of watery diarrhoea in otherwise healthy adults. ²	Adult: 2 mg TDS
Lomotil (diphenoxylate 2.5mg /atropine 25mcg)	Watch for opiate-induced ileus, drowsiness, and nausea caused by the atropine effects	Adult: 2 tablets TDS

	Effectiveness	Dose
Absorbent		
Kaolin mixture	Do not decrease the frequency or duration of diarrhoea and, thus, are not recommended in infectious diarrhea. ²	Child: 3 - 5 years: 5 - 10 ml TDS 6 - 12 years: 10 - 20 ml TDS > 12 years: 20 - 40 ml TDS Adult: 20 - 40 ml TDS
Diocetahedral Smectite (Smecta)		Child: > 2 years: 1 sachet BD - TDS
Charcoal		Adult: 400 mg TDS
Anti-spasmodic		
Simethicone drops (100mg/ml)	Relieve abdominal colic	Child: < 2 years or < 11 kg: 0.2 ml QDS ≥ 2 years or ≥ 11 kg: 0.4 ml QDS
Colimix (Dicyclomine 5 mg, Simethicone 50 mg per 5 ml)	Relieve abdominal colic	Child: 6 months - 4 years: 5 ml QDS 4 - 12 years: 5 - 10 ml QDS
Buscopan	Effective of abdominal cramps. Watch for anticholinergic, CNS, psychiatric side effects especially in older patients	Adult: 10 - 20 mg TDS
Intestinal secretion inhibitor		
Bismuth subsalicylate (Not available in NUP)	Decrease the frequency of diarrhoea and improve nausea and abdominal pain within 24 hours of treatment in acute diarrhoea patients ³	Adult: 2 tablets (262 mg/tab) or 30 ml (regular strength) ½-1hr Maximum daily dose: 8 regular-strength doses
Racecadotril (Hidrasec) (Not available in NUP)	Effective in diarrhoea in paediatric patients and has similar effects as loperamide in acute diarrhoea in adults. ^{4, 5}	Child: < 9 kg: 10 mg TDS 9 - <13 kg: 20 mg TDS ≥ 13 - 27 kg: 30 mg TDS ≥ 27 kg: 60 mg TDS Not recommended for < 3 months
Others		
Lactoguard	Limited evidence for probiotics as it has been difficult to interpret study results due to variability in probiotic strains and lack of dose-dependent effect. There have been more positive studies in children than adults that showed that probiotics were effective in decreasing duration and frequency of acute infectious diarrhea. ⁷ Nonetheless, probiotics are known to be safe with very little side effects.	Child: < 1 year: 1 sachet BD 1 - 10 years: 1 sachet TDS > 10 years: 2 sachet TDS Adult: 2 sachet TDS

3. Antimicrobial therapy

Most patients with acute gastroenteritis need not be treated with antimicrobial drugs. Acute watery diarrhoea is often viral in aetiology (norovirus, rotavirus, and adenovirus). Even when bacterial in aetiology, the symptoms often improve spontaneously without treatment, and treatment does not necessarily shorten the duration of symptoms. Considering the side effects and cost of antibiotics as well as antibiotic resistance, antibiotic treatment does not offer much benefit.⁸

Antimicrobial therapy can be considered in the following cases²:

- Blood or mucoid stools and fever
- Shigellosis symptoms (frequent scant bloody stools, fever, cramping abdominal pain, tenesmus)
- Traveller's diarrhoea with high fever > 38.5°C or sepsis as antimicrobial therapy shortens the course and alleviates the symptoms by 1.5 days⁸
- Bloody diarrhoea in immunocompromised patients

Empirical antibiotic therapy:

	First Line Therapy	Alternative Therapy
	Azithromycin 1000mg Q24H 1 day, OR 500mg Q24H 3 days	Ciprofloxacin 500mg Q12H 3 days
Use	¹⁰ Use empirically as first-line treatment for travellers' diarrhoea in Southeast Asia or other areas if fluoroquinolone-resistant bacteria are suspected. ¹¹ 24-hour dosing of azithromycin may be preferable to the 12-hourly dosing of ciprofloxacin	
Avoid	¹⁰ Side effects (mainly nausea) can limit the acceptability of single dose of 1,000 mg. Alternatively, use 500 mg a day for 3 days.	¹⁰ Increasing microbial resistance to fluoroquinolones among <i>Campylobacter</i> isolates. In South and Southeast Asia, where both <i>Campylobacter</i> infection and fluoroquinolone resistance are prevalent. ¹⁰ Increasing fluoroquinolone resistance has been reported from other destinations and in other bacterial pathogens, including in <i>Salmonella</i> and <i>Shigella</i> . ¹⁰ Fluoroquinolones carry a black box warning from the FDA regarding multiple adverse reactions, including aortic tears, hypoglycaemia, mental health side effects, and tendinitis and tendon rupture.

Special Situations

Specific antimicrobial therapy is given when a treatable pathogen is identified in stool cultures.^{1, 2}

Pathogen	First-line antibiotics	Second-line antibiotics
<i>Campylobacter</i>	Azithromycin	Ciprofloxacin
Non typhoidal <i>Salmonella</i>	Usually not indicated	NA
<i>Salmonella enterica</i> Typhi or Paratyphi	Ceftriaxone or ciprofloxacin	Ampicillin, TMP/SMX, Azithromycin
<i>Shigella</i>	Azithromycin, ciprofloxacin	TMP/SMX or ampicillin
<i>Vibrio cholerae</i>	Doxycycline	Ciprofloxacin, azithromycin
<i>Giardia lamblia</i>	Metronidazole	Tinidazole

Red Flags / Indications for Referral

Features present that may warrant further investigations or tertiary care:

1. Lethargy or confusion
2. Postural hypotension and tachycardia
3. Dehydration and inability to tolerate orally
4. Bloody stools
5. Temperature $\geq 38.5^{\circ}\text{C}$
6. Passage of ≥ 6 stools in 24 hours
7. Duration of illness > 72 hours
8. Severe abdominal pain in a patient > 50 years old
9. Diarrhoea in the elderly (≥ 70 years old)
10. Diarrhoea in the immuno-compromised
11. Chronic diarrhoea

Public Health

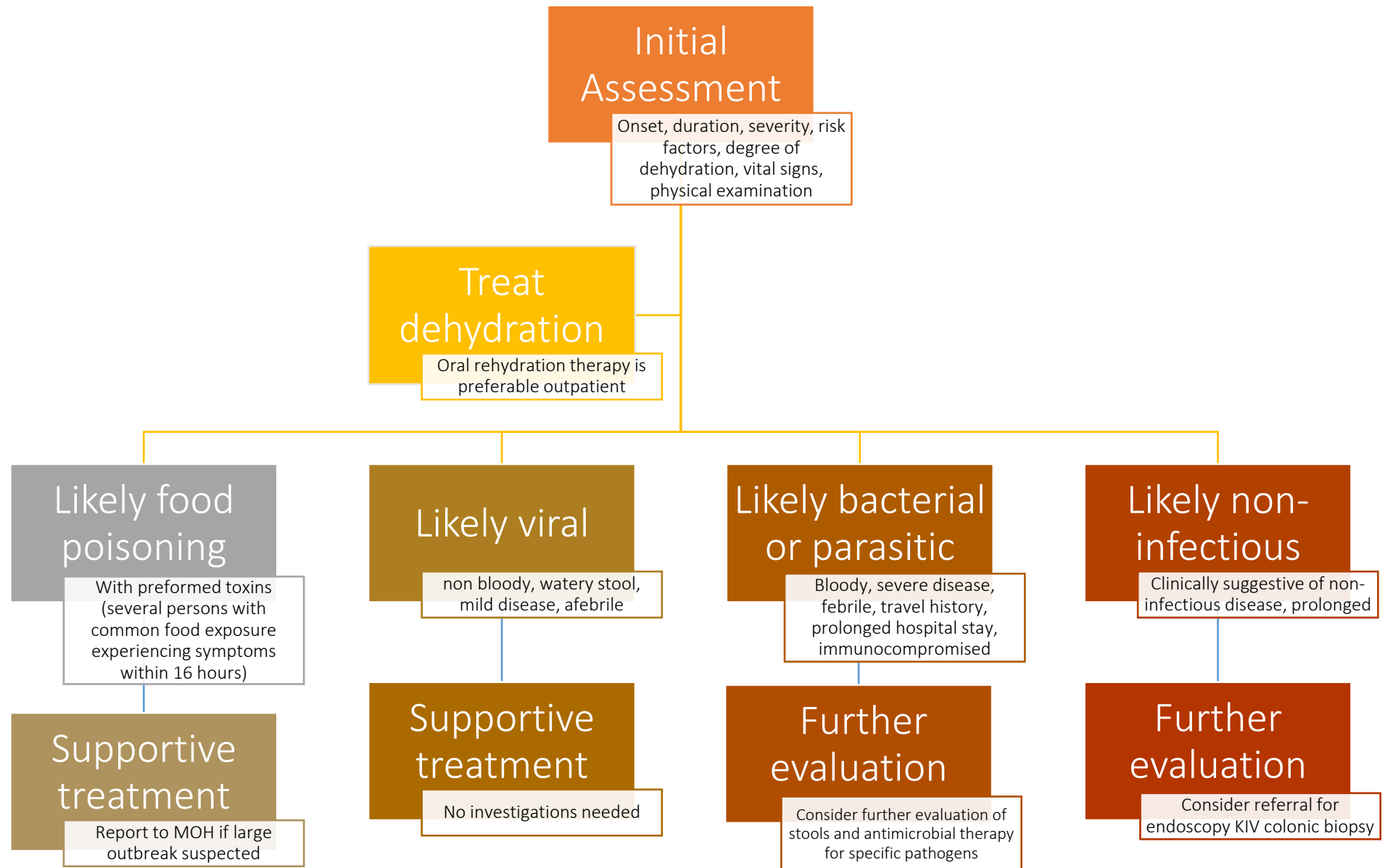
All suspected food poisoning outbreaks should be notified to the Ministry of Health to facilitate investigations (MD 131 or electronically via CD-LENS).

Key Recommendations⁹

<i>Clinical recommendation</i>	<i>Evidence rating</i>
The first step to treating acute diarrhoea is rehydration, preferably oral rehydration.	C
In patients with acute diarrhoea, stool cultures should be reserved for grossly bloody stool, severe dehydration, signs of inflammatory disease, symptoms lasting more than three to seven days, immunosuppression, and suspected nosocomial infections.	C
Antibiotics (usually a quinolone) reduce the duration and severity of traveller's diarrhoea.	A

A = consistent, good-quality patient-oriented evidence; B = inconsistent or limited-quality patient-oriented evidence; C = consensus, disease-oriented evidence, usual practice, expert opinion, or case series.

Summary



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